

Solution Architecture

Date 15 March 2026
Team ID xxxxxx
Project Name Smart agriculture production optimization engine
Maximum Marks 5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

[e.g., Web App (React), Mobile App (Flutter), Admin Dashboard]



API Gateway / Load Balancer

[e.g., Nginx, AWS API Gateway, Express Server]



Auth Service *[e.g., JWT, OAuth2]*

Core Logic Service

[e.g., Data Processing, Machine Learning Models]



External API Integrations

[e.g., Payment Gateway, Email Service]

Data / Storage Layer

[e.g., PostgreSQL Database, MongoDB, AWS S3]



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides the user interface where users enter weather details (rainfall, cloud visibility, seasonal rainfall, etc.) and view flood prediction results.</i>	<i>HTML, CSS, Bootstrap, Flask (Frontend Templates)</i>
<i>[API Gateway]</i>	<i>Receives user requests from the web application, validates them, and forwards them to the machine learning prediction service. Returns the prediction result to the user.</i>	<i>Flask REST API, Python</i>

<i>[Microservices]</i>	<i>Processes input data, loads the trained ML model (XGBoost), performs flood prediction, and sends the prediction result back to the API.</i>	<i>Python, Scikit-learn, XGBoost, NumPy, Pandas</i>
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*[Database] Stores historical weather data, user inputs (optional), and prediction records for future analysis and model improvement.
CSV Dataset,
SQLite (or MySQL)*